CS & DA

Probability and Statistics

Probability

Lecture No. 01







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Recap of previous lecture



Topic

PERMUTATIONS & COMBINATIONS

Topics to be Covered



Topic

BASICS of PROBABILITY (Part 01)

Q: In how many ways B'days of 6 different persons will fall in (i) Calendar Months (ii) In exactly 2 Calendar Months ? Ans: (66×62)

Sol: (i) Month = 12, Persons = 6 so Total ways of falling B'day = $\underline{12} \times \underline{12} \times \underline{12} \times \underline{12} \times \underline{12} \times \underline{12}$ $= (12^6)$ ways
$P_1 \quad P_2 \quad P_3 \quad P_4 \quad P_5 \quad P_6$
(No Restriction)

(ii) Months = 2, Persons = 6 so Required ways = ${}^{12}C_2$ ways $\times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2}$
$P_1 \quad P_2 \quad P_3 \quad P_4 \quad P_5 \quad P_6$
$= {}^{12}C_2 \times 2^6$ ways (But it is wrong)

So No. of ways in which 6 persons will take Birth in exactly 2 calendar months
(∵ out of 64 cases, 2 cases are unfavourable) $= {}^{12}C_2 \times (2^6 - 2) = 66 \times 62 = 4092$ ways.

Analysis: i.e No. of ways of selecting 2 months $= {}^{12}C_2 = 66$ ways.

(J-F), (J-M), (J-A), (J-M) ..... (J Nov), (J Dec) = 11 ways

(Feb M) (FA) (FM) ---- (F Nov) (F Dec) = 10 ways

(MA) (MM) (MJ) ------ (M Nov) (M Dec) = 9 ways

(Oct Nov), (Oct Dec) ⟶ = 2 ways

(Nov Dec) ⟶ = 1 way

+

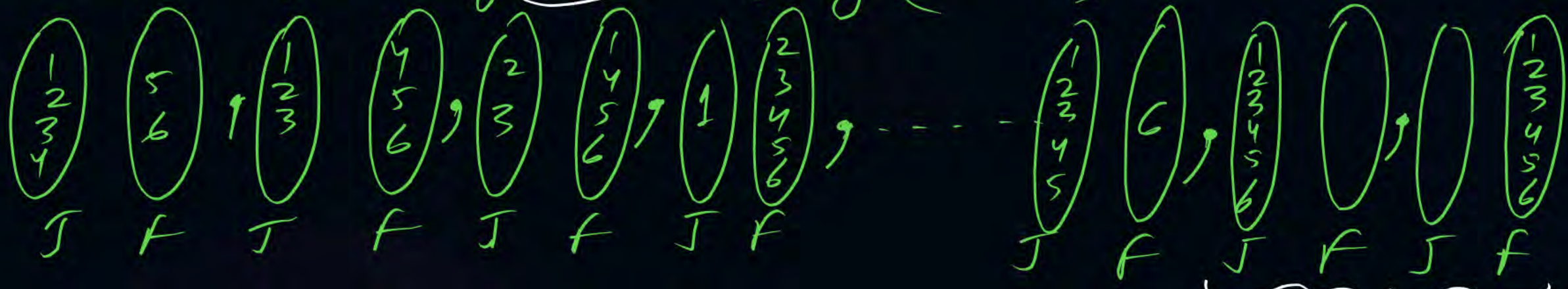
66 ways.

$S_n = \frac{n(n+1)}{2}$

$S_{11} = \frac{11 \times (11+1)}{2}$

$= 66$ ways

Let up take one pair out of 66 pair ie for eg (Jan & Feb)



64 Cases

So fav Cases for (Jan & Feb) = 64 − 2 = 62

Hence Req Ans = $\underbrace{62+62+62+62+\cdots+62}_{66 \text{ Cases}} = 66\times62 = 4092$ Cases

PROBABILITY (possibility) / chance

Random Exp.: → when outcome of an Exp is not sure then such types of Exp are called R.Exp for eg Tossing a Coin, Throwing a Die etc

Sample space (S) → various possible outcomes written in a set form is called S.space

$S_{coin} = \{H, T\}$, $S_{Die} = \{1, 2, 3, 4, 5, 6\}$, $S_{family} = \{B, G, T\}$

Event (E) → Any subset of S is called an Event

$S = \{1, 2, 3, 4, 5, 6\}$

- $E_1 = \{1, 3, 5\} = \{$odd No occurs$\}$
- $E_2 = \{2, 4, 6\} = \{$Even No ,, $\}$
- $E_3 = \{1, 2, 3, 4\} = \{$No ≤ 4 occurs$\}$
- & so on.....

Impossible Event → $\because \phi \subset S$ so ϕ is also an Event & it is called Impossible Event.

& $\boxed{P(\phi)=0}$ for eg for Die; $E=\{\text{getting -ve No}\}$ then $E=\phi$

Sure Event / Certain Event → $\because S \subseteq S$ so S is also an Event & it is called sure Event

& $\boxed{P(S)=1}$ for eg $E=\{1 \le \text{No} \le 6\}=\{1,2,3,4,5,6\}$ so $E=S$

Note →

① $0 \le P(E) \le 1$, ② $P(\text{Something occurs})=1$, ③ $P(\text{Nothing occurs})=0$

④ $P(\text{given statement})=1$, ⑤ $P(\text{Death})=1$, ⑥ $P(\text{GOD})=1$

⑦ Prob $\simeq$ Base is of 1 unit | Proportion = Base is of again 1 unit

% $\simeq$ " " " 100 unit

$M:W=3:4$ → Prop of $M=\frac{3}{3+4}=\frac{3}{7}=\frac{3/7}{1}$

Prop of $W=\frac{4}{3+4}=\frac{4}{7}=\frac{4/7}{1}$

⊛ If sample size is small then Prob ≅ Proportion

⊛ Total Number of subsets of any set having n different elements $= 2^n$

Hence Total Number of Events associated with sample space 'S' = Total No of subsets of S $= 2^n$

eg $S = \{1, 2, 3\}$ so Total No of subsets $= 2^3 = 8$

Various subsets: $\{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}$ (Proper subsets), $\phi, \{1,2,3\}$ (Improper subsets)

Note:- $\phi = \{\}$ = Null set / Void set / Empty set

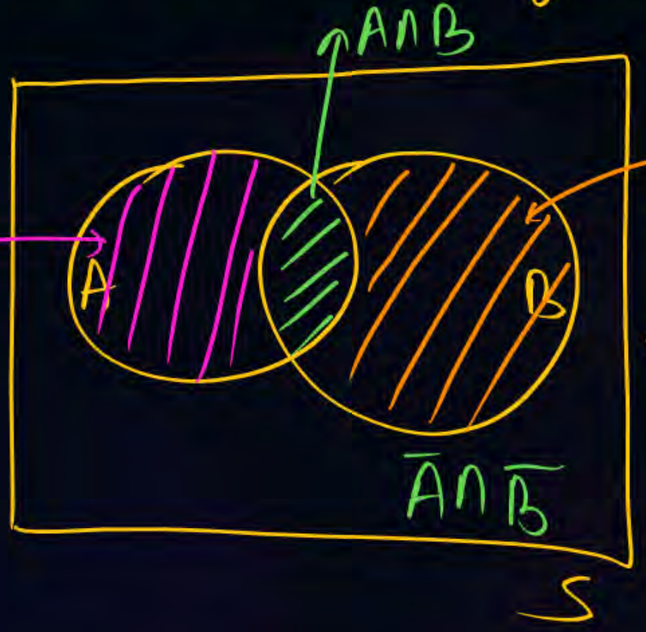
(*) Either A or B or Both = Atleast one of A or B = $A \cup B$

Both A & B = Simultaneous occurrence of A & B = $A \cap B$

Neither A Nor B = None of A & B = $\bar{A} \cap \bar{B}$

} Imp.

(*)



① (only A) or (only B) or (Both A & B) = Either A or B

$$(A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (A \cap B) = A \cup B$$

② $A \cap \bar{B} = A - (A \cap B)$, $B \cap \bar{A} = B - (A \cap B)$

③ (Either A or B) + (Neither A Nor B) = S.Space

ie Neither A Nor B = S − Either A or B

$$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$$

(4) W.K.T (Either A or B) + (Neither A nor B) = S.Space

(At least one of A or B) + (None of A & B) = Total (S)

$$P(\text{at least one}) = 1 - P(\text{None})$$

(5) $P(A \cup B) = P[(A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (A \cap B)]$ = very lengthy

Shortcut :- Addition Th of Prob :- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Addition Th for three Events :- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

(6) Multiplication Th of Prob :- $P(A \cap B) = P(A/B) \cdot P(B)$

Independent Events → Events are called (Ind) if occurrence or Non occurrence of one Event do not alter the occurrence or Non occurrence of other Events

"In Case of Ind. Events, we can Multiply the respective probabilities in order to find their Simultaneous prob"

if A & B are (Ind) $\iff$ $\boxed{P(A \cap B) = P(A) \cdot P(B)}$

Eg $S_C = \{H, T\}$, $S_D = \{1, 2, 3, 4, 5, 6\}$

$A = \{H\}$, $B = \{No \leq 4\}$

$P(A) = \frac{1}{2}$, $P(B) = \frac{4}{6}$

Here A & B are Ind. (By Common Sense)

$\Rightarrow P(A \cap B) = ? = P(A) \cdot P(B)$

$= \frac{1}{2} \times \frac{4}{6}$

Note:- Here $A \cap B = \phi$ But this Concept is Meaningless in Case of Ind. Events.

⊛ Mutually Exclusive Events

"If two Events can't occur simultaneously then Events are called ME Events" or

If occurrence of one event prevents the occurrence of other Events then Events are called ME Events. ie only one Event at a time can occur.

A & B are called ME if $A \cap B = \phi$ ⟺

- $P(A \cap B) = 0$
- $P(A \cup B) = P(A) + P(B) - 0$

eg: $S_{dice} = \{1,2,3,4,5,6\}$, $E_1 = \{1,3,5\}$, $E_2 = \{2,4,6\}$, $E_3 = \{1,2,3,4\}$

E_1 & E_2	E_2 & E_3	E_1 & E_3
$\because E_1 \cap E_2 = \phi$	$\because E_2 \cap E_3 \neq \phi$	$\because E_1 \cap E_3 \neq \phi$
Hence ME	Not ME	Not ME
$P(E_1 \cap E_2) = 0$		

Ex) V.V.Imp Conclusions

Ind. Events are associated with Different sample spaces while

M.E Events " " with same sample space.

Note:

$2 = 2$ obvious

$2^2 = 2^2$

$(2^2 - 2^2) = 0$

$(2-2)(2+2) = 0$

$0 \times 4 = 0$

$4 = \frac{0}{0}$

i.e we should avoid to draw conclusions from obvious statements.

(Imp) Nature of Elements in S Space → If R.Exp is repeated n times then elements of S.space are in the form of ordered n-tuple.

eg, If a Die is Rolled (once) then $S=\{1,2,3,4,5,6\}$ & $n(S)=6^1$

eg If a Die is thrown (twice) then $S=\begin{Bmatrix}(11),(12),(13),(14),(15),(16)\\(21)(22)\quad\cdots\quad(26)\\(31)(32)\quad\cdots\quad(36)\\\vdots\qquad\qquad\vdots\\(61),(62)\quad\cdots\quad(66)\end{Bmatrix}\Rightarrow n(S)=\underset{n_1}{6}\times\underset{n_2}{6}=36$ 36 O-pair

eg If a Coin is tossed (5 times) then $S=\{(HHHHH),(HHHHT),(HHHTT),\ldots,(TTTTT)\}\Rightarrow n(S)=\underset{C_1}{2}\times\underset{C_2}{2}\times\underset{C_3}{2}\times\underset{C_4}{2}\times\underset{C_5}{2}$ $=32$ ordered 5-tuples

eg A Couple has 3 children then $S=\begin{Bmatrix}(BBB),(BBG),(BGB),(BGG)\\(GBB),(GBG),(GGB),(GGG)\end{Bmatrix}\Rightarrow n(S)=\underset{C_1}{2}\times\underset{C_2}{2}\times\underset{C_3}{2}=8$ Triplets.

Methods of Solving Questions →

Approach 1 → By Making S.space(S) & Fav Event(E) in terms of set, Req Prob will be

$$P(E) = \frac{n(E)}{n(S)}$$

App(2) → when it is not easy to write S & E in terms of set then directly find Fav Number of Cases & Total No of Cases by using the concept of P&C

& then Req Prob $= \frac{\text{fav No. of Cases}}{\text{Total No. of Cases}}$

App(3) → By using some Standard Def^{n} & Standard Results

Whenever given Information is in the form of Prob then we should use App 3.

(*) Fav Cases :→ Fav = { which is Required & should be assumed as Fav }

(*) Revision : = { Desired outcome " " " " }

Operation	P ∈ C	Prob	ME	Ind
Either or	Add	∪ use Addition Th	$P(A \cup B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$
AND	Multiply	∩ use Multi Th	$P(A \cap B) = 0$	$P(A \cap B) = P(A) \cdot P(B)$

(*) (1) Three Dice are thrown simultaneously
(2) A Die is thrown three times

} Both have same S space

$S = \{(111)(112)\ldots(116), (211)(212)\ldots(666)\} \Rightarrow n(S) = 6 \times 6 \times 6 = 216$ Triplets

Q: Four Dices are thrown simultaneously then find the prob that sum of the outcomes will be 22?

Sol: $S = \{(1111), (1112), \ldots, (1116), (2111), (2112) \ldots\ldots, (6666)\}$

$$n(S) = \underset{D_1}{6} \times \underset{D_2}{6} \times \underset{D_3}{6} \times \underset{D_4}{6} = 6^4 = 1296 \text{ Quadruples}$$

App① fav Event $(E) = \{\text{sum is } 22\} = \{(6664), (6646), (6466), (4666), (6655), (6565), (6556), (5566), (5656), (5665)\} \Rightarrow n(E) = 10$ Quadruples

So Req Prob $P(E) = \frac{n(E)}{n(S)} = \frac{10}{1296}$

App② fav Quadruples $= \{\text{eg } (6664)\ldots\ldots, \text{eg } (6655)\ldots\ldots\}$ $\rightarrow \frac{4!}{3!} = 4$ or $\frac{4!}{2!2!} = 6$

i.e total 10 Quadruples are fav

So Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{10}{1296}$

HW8 7 surgical strikes occurred in a week from INDIA on PAKISTAN then find the prob that all will occur on a same day ?

Abk ② S. Strikes → Different

Ans = $\frac{1}{7^6}$

THANK - YOU